

Module 5.1: HBase

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Distributed Databases

A distributed database is a database whose storage devices are not attached to a common processing unit but are distributed across a network. It is a collection of logically related data that is physically stored at different locations and managed by a Distributed Database Management System (DDBMS). Distributed databases are designed to improve scalability, reliability, availability, and performance in large-scale applications. In modern computing environments, organizations generate massive amounts of data that cannot be efficiently handled by a single database server. Distributed databases solve this problem by distributing data across multiple nodes while allowing users to access and manage data transparently.

1. Data is stored on multiple machines.
2. Users view the database as a single system.
3. Data can be accessed from different locations.
4. Supports data sharing and collaboration.
5. Managed by a Distributed Database Management System (DDBMS).

Example: A multinational company may store employee records in different countries while allowing centralized access through a distributed database system.

Characteristics of Distributed Databases

- **Data Distribution** – Data is stored across multiple locations or nodes.
- **Transparency** – Users access data without knowing its physical location.
- **Scalability** – The system can grow by adding more nodes.
- **Reliability** – Data remains available even if some nodes fail.
- **Concurrency** – Multiple users can access and modify data simultaneously.
- **Autonomy** – Each site can operate independently while remaining connected.

Advantages of Distributed Databases

- **High Availability** – Services remain operational despite node failures.
- **Improved Performance** – Data can be processed closer to where it is stored.
- **Scalability** – Additional nodes can be added to handle increased workloads.
- **Fault Tolerance** – Data replication helps prevent data loss.
- **Reduced Communication Cost** – Local access reduces network traffic.
- **Flexibility** – Supports geographically distributed users and applications.

Challenges of Distributed Databases

- **Data Consistency** – Keeping replicated data synchronized is difficult.
- **Complexity** – Management and administration are more complicated.
- **Security Issues** – Protecting data across multiple locations is challenging.
- **Network Dependence** – Performance depends on network reliability.
- **Transaction Management** – Coordinating transactions across nodes is complex.
- **Higher Cost** – Additional hardware, software, and maintenance may be required.

Distributed Databases in Big Data

Big Data applications generate enormous volumes of structured, semi-structured, and unstructured data. Traditional databases often struggle to manage such massive datasets. Distributed databases play a crucial role in Big Data environments by providing scalable and fault-tolerant storage solutions.

Role in Big Data

1. Store petabytes of data across multiple nodes.
2. Support parallel data processing.
3. Provide high availability and fault tolerance.
4. Enable real-time and batch analytics.
5. Handle rapidly growing datasets efficiently.

Popular Distributed Databases in Big Data

Database	Type
HBase	Column-Oriented NoSQL Database
Cassandra	Distributed Column-Family Database
MongoDB	Document-Oriented Database
Couchbase	Distributed NoSQL Database

Applications

1. Social media platforms.
2. E-commerce websites.
3. Banking and financial systems.
4. Healthcare data management.
5. IoT and sensor networks.
6. Cloud computing environments.

Importance: Distributed databases form the foundation of modern Big Data ecosystems such as Hadoop and Spark. They provide scalable storage, fault tolerance, and high-speed data access required for processing massive datasets efficiently.

HBase

HBase (Hadoop Database) is an open-source, distributed, column-oriented NoSQL database built on top of HDFS. It is designed to store and manage large volumes of structured and semi-structured data across clusters of commodity hardware. HBase provides real-time read and write access to big data and is modeled after Google's Bigtable.

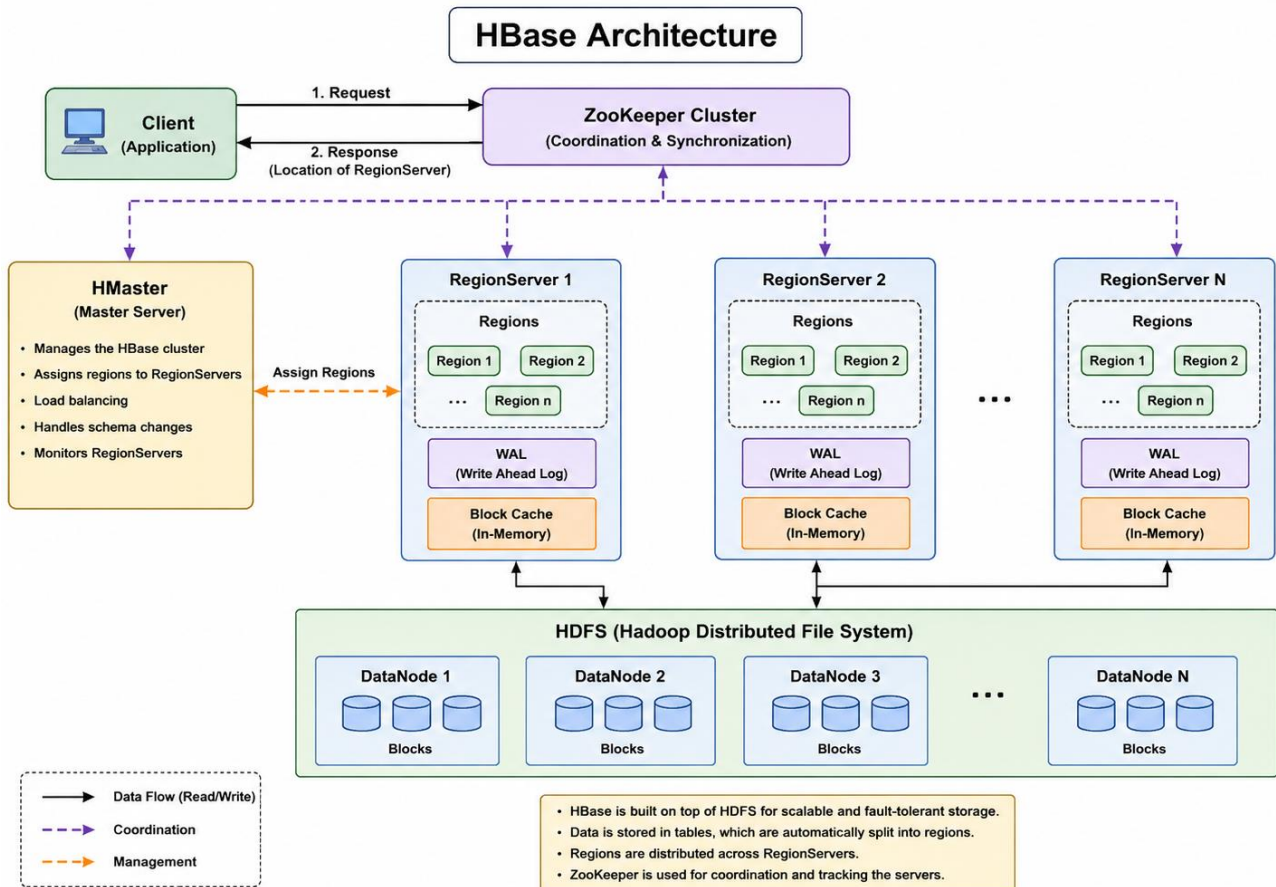
- Runs on top of HDFS.
- Supports random, real-time read/write operations.
- Designed for massive scalability.
- Suitable for sparse and large datasets.
- Integrates well with Hadoop ecosystem tools.

Features of HBase

- **Distributed Storage** – Data is stored across multiple nodes in a cluster.
- **Scalability** – Can scale horizontally by adding more servers.
- **Fault Tolerance** – Uses HDFS replication for data reliability.
- **Real-Time Access** – Supports fast read and write operations.
- **Column-Oriented Storage** – Data is stored by columns instead of rows.
- **Strong Consistency** – Ensures consistent data reads and writes.
- **Flexible Schema** – Columns can be added dynamically.
- **Integration with Hadoop** – Works seamlessly with HDFS, MapReduce, and YARN.

Architecture of HBase

HBase follows a master-slave architecture where a central HMaster coordinates multiple RegionServers. Data is stored in HDFS, while ZooKeeper manages coordination and monitoring.



- 1. HMaster:** HMaster is the master server responsible for managing and coordinating the HBase cluster.
 - Assigns regions to RegionServers.
 - Performs load balancing.
 - Handles schema changes.
 - Monitors RegionServer status.
 - Recovers from server failures.
- 2. RegionServer:** RegionServer stores and manages table regions and serves client requests.
 - Handles read and write operations.
 - Manages table regions.
 - Communicates with HDFS.
 - Splits large regions automatically.
 - Maintains in-memory cache for faster access.
- 3. ZooKeeper:** ZooKeeper is a distributed coordination service used by HBase.
 - Tracks active servers.
 - Maintains cluster configuration.
 - Helps clients locate RegionServers.
 - Detects failures.
 - Coordinates distributed operations.
- 4. HDFS:** HDFS is the storage layer of HBase.
 - Stores HBase data files.
 - Provides fault tolerance through replication.
 - Ensures high availability.
 - Supports distributed storage.
 - Handles large-scale data efficiently.

Working

1. Client sends request to HBase.
2. ZooKeeper identifies the appropriate RegionServer.

3. RegionServer processes the request.
4. Data is stored or retrieved from HDFS.
5. HMaster manages load balancing and region assignments.

HBase Data Model

HBase uses a column-oriented data model based on Google's Bigtable. Data is organized into tables, rows, column families, columns, and cells. **Components of HBase Data Model**

1) Table

- A collection of related data.
- Similar to a table in a relational database.
- Consists of rows and column families.

2) Row Key

- Unique identifier for each row.
- Rows are stored in sorted order based on row keys.
- Used for fast data retrieval.

3) Column Family

- Groups related columns together.
- Defined during table creation.

- Physical storage unit in HBase.

4) Column Qualifier

- Specific column within a column family.
- Can be added dynamically.

5) Cell

- Intersection of a row and column.
- Stores the actual data value.

6) Timestamp

- Each cell contains a timestamp.
- Allows storage of multiple versions of data.

Table: Student

Row Key	Column Family: Info	Column Family: Marks
101	Name = John Age = 20	Java = 85 DBMS = 90
102	Name = Alice Age = 21	Java = 88 DBMS = 92

HBase Implementation

HBase is implemented on top of HDFS and follows a distributed architecture.

Working

1. Data is divided into regions.
2. Regions are distributed across RegionServers.
3. RegionServers store data in HDFS.
4. HMaster manages the cluster.
5. ZooKeeper coordinates communication.

6. Clients access data through RegionServers.

Advantages of Implementation

- Horizontal scalability.
- High availability.
- Fault tolerance.
- Real-time data access.
- Efficient handling of large datasets.

HBase Clients

HBase clients are applications or APIs used to interact with HBase tables. **Types of HBase Clients**

1) Java API Client

- Most commonly used client.
- Provides direct access to HBase operations.
- Supports CRUD operations.

2) HBase Shell

- Command-line interface.
- Used for administration and testing.
- Supports table creation and data manipulation.

3) REST Client

- Accesses HBase through REST APIs.
- Suitable for web applications.

4) Thrift Client

- Provides cross-language access.
- Supports Python, C++, PHP, Ruby, etc.

5) MapReduce Client

- Integrates HBase with Hadoop MapReduce.
- Used for large-scale data processing.

Practical Applications of HBase

1. **Social Media** – Stores user profiles, posts, comments, and activity logs.
2. **E-Commerce** – Manages product catalogs, customer information, and purchase history.
3. **IoT Systems** – Stores continuous streams of sensor-generated data.
4. **Banking and Finance** – Maintains transaction records and customer account data.
5. **Healthcare** – Stores patient records, prescriptions, and medical histories.
6. **Telecommunications** – Manages call records and network usage data.
7. **Web Analytics** – Tracks website traffic, user behavior, and clickstream data.
8. **Recommendation Systems** – Stores user preferences for personalized recommendations.
9. **Fraud Detection** – Processes large datasets to identify suspicious activities.
10. **Online Gaming** – Maintains player profiles, scores, and game statistics.